

16/07/24:-

CER 3G1 → computer organization and Architecture

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Computer

↳ Hardware/
Software

↳ Need program/
software to
perform calc.

↳ It is a programmable
machine.

Calculator

↳ Hardware/
firmware

↳ don't need
any software

↳ It is a simple
Hardware Machine
with some limitations

↳ All computer → Same design → different components

↳ i3 differ from i7 → Because of Hardware present →
increases the ability & analyzing Power

↳ Program/code error → syntax error

↳ Computer Memories/ → ① RAM ② Pendrive
storage
↳ 8GB & 16GB
↳ 32GB & 64GB

30/July/24:-

Computer Organization & Architecture:- (Book name)
By William Stallings (10th ed.)
(suferred Book)

◎ Boolean → Powerful mathematics, which gives the evolution of digital system. (in Binary)

machine language (in Binary format)
Assembly language
~~XXXXXX~~ C
4GLS

} processor dependent

} processor independent
as we go downwards,
coding implemented in
english language

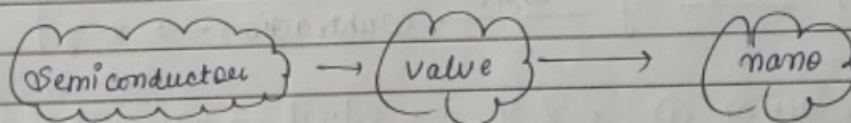
(HDD → Hardware device) ← PC

SSD (Semiconductor device)
mechanical devices

Both are memory organizer

But SSD is faster 'cause of Semiconductor

↳ Architecture is the logical part, how does it works (organiser)
↳ organizer is a organised part



↳ use of biochips → No connectivity, but it transform signals. (it will come in future)

↳ certain chemical present → Brain → perform → Logic gates

↳ Logic gates used as a building block while studying mathematical de-morgan's law, Arithmetic op.
↳ +, -, *, ÷

→ using X OR GATE $\xrightarrow{\text{perform}}$ 2 operation $\rightarrow +, -$

→ By using repetitive Addition, we can $\rightarrow *$ perform

→ " " " subtraction " $\rightarrow \div$

{ Basic architecture $\rightarrow$ particularly organized $\rightarrow$ we can designed $\rightarrow$ a perfect device }

ALU $\rightarrow$ Arithmetic Logical unit

if } $\rightarrow$ single statement $\rightarrow$ compare & Jump
else }

cmp (instruction)

$\begin{matrix} < > < > \\ \downarrow & \downarrow \\ \text{Register ①} & \text{Register ②} \end{matrix}$ } $\rightarrow$ compare Register ① & Register ②

Accumulate

$\left[x \xrightarrow{\text{compare}} y \xrightarrow{\text{Logic}} \text{Subtraction} \rightarrow x - y \right]$

→ condition {
① $x \& y$ are equal $\rightarrow$ result = 0
② $x > y \rightarrow$ result (+)
③ $x < y \rightarrow$ result (-)

part of Architecture

→ CPU used in computers $\rightarrow$ intel processor
→ mac processor } $\rightarrow$ sudden partition must be differ.

- Architectural difference
- Organisational difference

→ in intel & mac processor

mac → RISC (Reduced Instruction set computer)
 intel → CISC

RISC → Less NO. of instructions

CISC → more NO. of instructions

→ Android using ARM Processors
 Mobile users

→ peripherals, i/o devices

→ How they were connected
 → " " " used

In systems:-

[power supply doesn't affect the performance → power supply (Mechanical device)]

KB → mechanical device

then, Hard-disk, SSD

→ Instruction cycle

- Types of Instruction
- How Instruction works
- What is Instruction

Study

→ Optimization is always required

→ in contrast of Speed and technology

* Always used variable → acc-to your compiler
 memory

memory → 0mb — 64kb

#Aug/24:-

ch-1, 2

computer Architecture & organization:-

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value System

stored program

→ Functions

uses of computers:-

- Data Processing
- Data Storage
- " Movement
- Control

→ In programming,
we have done all of
these four functions

→ Many times retrieve (data) → Permanent storage

→ Memory → 2 types → volatile memory (RAM)
→ Permanent " (Floppy disk)

→ Talking about → Hardware
Structure

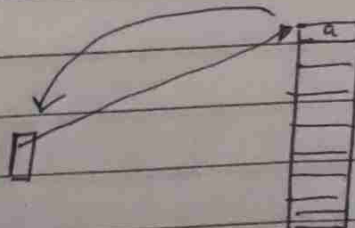
ISA

→ microprogrammed → have some structure → imposed some architecture → How it designed

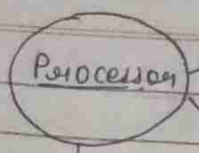
→ without this concept,
we are unable to
make ALU, Bus structure etc

→ fetch cycle → Access bus } Fig 1.8 in (CHAPTER-1)
→ execution cycle → No work on bus }

- Fetch part will access your data bus
- ALU will execute the instructions



→ Fetch → bring it → CPU from memory → to decode (execution part)



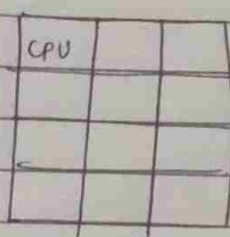
single processor

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Multi processor → it may have, more than one processors, more than one processing unit → called → CORE

the physical device

(CPU) is the part of the processor



single

→ on a same device, there are several units

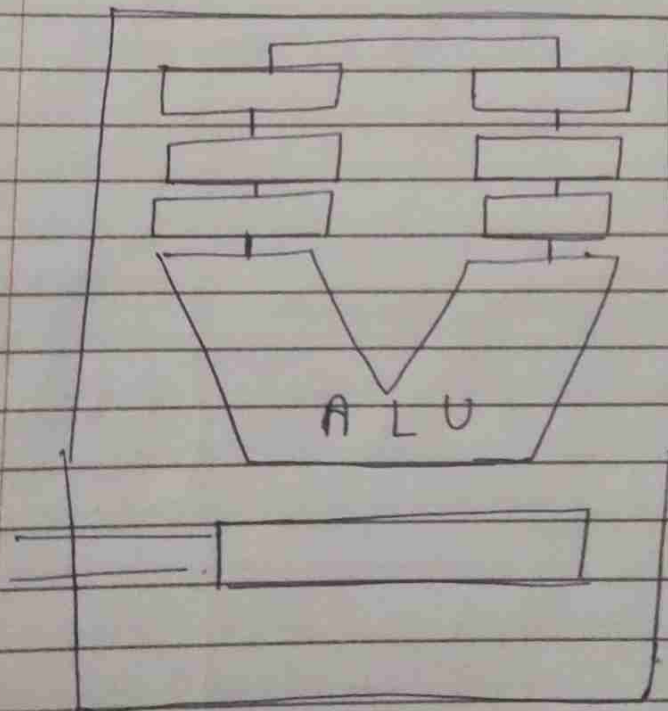
→ Today we are using multi-core processor

A single processor → it must have — ① ~~ALU~~ Registers (several registers)

② ALU

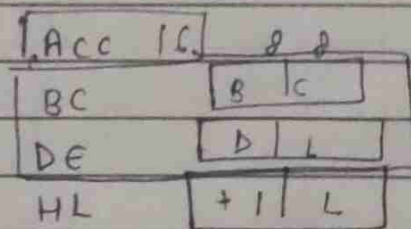
③ Control unit

④ CPU Inter connection



*Exc

X86



PC — 16 bit
 SP — 16 bit

Point always

whenever we use JUMP

POP
 → PUSH

used

$f(\&\text{int}; \&\text{int})$
 $((\text{int}^*; \text{int}^*))$

$f(x, y)$
1

$f(\&x, \&y)$